

IMMIGRANT WORKERS AND FARM PERFORMANCE: EVIDENCE FROM MATCHED EMPLOYER-EMPLOYEE DATA

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In many developed countries, the agricultural sector has experienced a significant inflow of immigrants. At the same time, agriculture is still in a process of structural transformation, resulting in fewer but larger and presumably more efficient farms. We exploit matched employer-employee data for Danish farms in 1980–2008 to analyze the micro-level relationship between these two developments. Farms employing immigrants tend to be both larger than and no less productive than other farms. Furthermore, an increased use of immigrants is associated with an improvement in job creation and revenue, which at least partially seems to reflect a causal effect of immigrants.

Key words: immigrants, job creation, productivity, revenue.

JEL codes: F22, Q12.

In many developed countries, the agricultural sector—or the rural sector more generally—has employed an increasing number of immigrants in recent years. According to the National Agricultural Workers Survey (NAWS), the number of foreign-born newcomers in the U.S. farm workforce, i.e., immigrants who have been in the country for less than one year, increased from 10% in 1993–1994 to 16% in 2001–2002.¹ In Europe, countries like Italy, Spain, Portugal, and Greece have also experienced a rapid increase in migrant employment in the agricultural sector during recent decades (Kasimis 2005), and in Denmark the share of immigrant workers in the agricultural sector increased from less than 2% in 1993 to almost 13% in 2008.

Furthermore, agricultural sectors worldwide are still in a process of structural transformation where many farms close, while others grow larger and presumably more productive. In the United States, for example, the number of farms decreased from around 6 million in the first half of the twentieth century to around 2 million by the beginning of the twenty-first century. In the same period, average U.S. farm size more than doubled, while agriculture's share in total employment fell dramatically.² As will be shown, similar developments have taken place in Denmark.

It is generally believed that migrant labor has helped to fill labor deficits and has reduced labor costs in agriculture (e.g., Kasimis 2005; Huffman 2005), but the question is whether there is also a more direct micro-level relationship between the use of immigrants and the structural transformation of farms. In other words, does the employment of immigrants help a farm to expand and/or become more productive? Or is it only those farms that grow and succeed (or the farms that do not) that are able to hire/use immigrants? Or is there no relationship between these two

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¹ Information about the National Agricultural Workers Survey (NAWS) can be found at www.doleta.gov/agworker/naws.cfm.

² The data are from the United States Department of Agriculture. www.usda.gov.

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developments at the farm level? The purpose of this article is to take a first step towards answering these questions. This, however, requires comprehensive farm-level data about the composition of employment at the individual farm, and such data are rarely available. Fortunately, we have access to a unique linked employer-employee dataset for the entire population of farm establishments and workers in the Danish agricultural sector over almost a 30-year period, from 1980 to 2008.

Theoretically, we can imagine at least three different situations. First, immigrants could simply represent an aggregate supply shift, with immigrants taking up a larger share of the total labor supply in the agricultural sector, but with no farm-level relationship between the employment of immigrants and farm performance. Second, immigrants may constitute a cheap and/or more flexible source of labor that benefits the individual farm. Other studies indicate that immigrants receive lower wages; this is the case both within agriculture and within other sectors of the economy (e.g., Card 2005). While part of the lower wage may reflect a lower effective labor input of the immigrant, part of it is also likely to reflect more limited outside options for immigrant workers, such as limited employment opportunities in other sectors of the economy, making them a cheaper source of labor (e.g., Malchow-Møller, Munch, and Skaksen 2012). This could in turn improve performance, as measured by, for example, profit and scale, for farms with access to immigrant labor.³ Third, it could also be the case that only certain types of farms are able or willing to hire immigrants and that these are also the more (or less) successful farms. The existence of fixed and sunk costs of hiring immigrants (a change of working language, attitudes, etc.), which may vary across farms, and/or the fact that immigrants cannot manage all functions at a farm, may imply that it requires a farm of a certain type or size (or with a certain location) to employ immigrant labor, and these farms may at the same time perform better (or worse) than other farms. In the case of such selection effects, the causality runs from (unobserved) farm characteristics to the employment of immigrants.

The second and third possibilities are, of course, not mutually exclusive. It could well be that immigrants are only hired by certain types of farms and that they subsequently affect the performance of these farms. The purpose of the present article is to provide some initial evidence on the importance of these different possibilities.

Specifically, we ask the following three questions in the article: Are farms that employ immigrants different from other farms? What is the relationship between an intensive use of immigrants and farm performance? Does this reflect a causal effect of immigrants?

We answer these questions in the following way. First, we set up a simple theoretical model to illustrate the three abovementioned theoretical possibilities. The model is used to guide the subsequent empirical analysis. Second, we compare key characteristics of farms that employ immigrants to farms that do not employ immigrants to establish whether any differences exist. To the best of our knowledge, this has not been done systematically before. Third, we estimate the relationship between the employment of immigrants and farm performance. We use several different measures of farm performance, including job creation, survival, revenue, and labor productivity. To control for potential endogeneity of the immigrant variable, we use regressions in first differences as well as instrumental variables (IV). This allows us to take a first step towards disentangling any causal effects of immigrants from pure selection effects.

To preview our results, we find that farms employing immigrant workers are larger in terms of employees, revenue, and value added, and are at least as productive as other farms. Furthermore, an increased use of immigrants—especially immigrants from Eastern Europe and less developed countries (LDCs)—is associated with increases in total employment and revenue. Our IV estimations indicate that at least some of these increases reflect a causal effect of immigrants. However, the expansionary effects are quantitatively modest and short-lived, possibly because the persistence in the use of immigrant labor on the individual farm is low. As the effects are strongest for the lowest-paid immigrants, and since native employment (especially unskilled native employment) is reduced and labor productivity remains unaffected by the employment of immigrants, our findings seem to support the idea that immigrants constitute a cheaper source of labor that can substitute for

³ Devadoss and Luckstead (2008) have argued that immigrants may also provide complementary input to capital and educated (native) labor. This could be in the form of, for example, knowledge about different production techniques and foreign markets.

(unskilled) native workers, thereby reducing the marginal cost of labor for farms and allowing them to expand.

Several studies have been done on the more aggregate/general effects of immigration, particularly the effects on wages and employment of native workers. Most of these analyses focus on the wage or net employment consequences for groups of individuals (regions, industries, or skill groups) following an increase in the supply of immigrant workers. Examples of such analyses are Card (1990, 2001, 2005); Borjas, Freeman, and Katz (1997); Pischke and Velling (1997); Borjas (2003, 2006); Angrist and Kugler (2003); Dustmann, Fabbri, and Preston (2005); Ottaviano and Peri (2012); and Aydemir and Borjas (2007). Substantial variation exists in the findings of these studies. In Longhi, Nijkamp, and Poot (2005, 2006), the results from a large number of analyses are compared, and the authors conclude that in general immigration has a small negative employment and/or wage effect for native workers.

Within agricultural economics, increasing attention has been given to the importance of immigrant workers, and Partridge, Rickman, and Ali (2008) argue for potentially different effects of immigrants in this sector than in the rest of the economy. However, the number of studies focusing explicitly on the agricultural sector is much more limited, and existing studies of immigrants in agriculture have typically relied on either aggregate data or relatively small household samples, which do not allow them to address the above issues (e.g., Taylor and Martin 1997, 2003; Devadoss and Luckstead 2008).

By estimating a simultaneous equations model on data from Californian towns, Taylor and Martin (1997) find evidence of a cyclical relationship between immigration and farm employment. An increase in the number of immigrants increases farm employment, and an increase in farm employment also raises immigration. A similar finding is reported in Taylor and Martin (2003). For a summary of these findings, see Martin and Taylor (1998) or Taylor and Martin (2001).

Based on a calibration exercise, Devadoss and Luckstead (2008) have recently argued that an increase in the use of immigrant workers in vegetable production in California actually has a very small negative effect on native employment in that sector. Related to this, Venturini (1999) finds some evidence of displacement, as an increase in the number

of illegal immigrants working in the Italian agricultural sector is found to reduce the number of natives employed in the sector.

The rest of the article is structured as follows. In the next section, we present our theoretical model. Based on this, we lay out the empirical strategy. We then describe the data used in the article and present some descriptive statistics before we turn to the results of the empirical analysis. The final section concludes.

Theory

If immigrants are identical to native workers, an increase in the number of immigrant workers in an agricultural sector will be similar to a general labor supply increase. In this case, we would expect no farm-level relationship between the use of immigrants and performance. The purpose of this section, however, is to illustrate why the use of immigrants and farm performance may be correlated, and how this may reflect a causal effect of immigrants and/or farm-specific factors that influence both the use of immigrants and farm performance.

To illustrate the theoretical possibilities, let profit by farm i be given by:

$$(1) \quad \pi_i = f_i(L_i^N, L_i^F) - C^N(L_i^N) - C_i^F(L_i^F),$$

where $f_i(L_i^N, L_i^F)$ is the production function of farm i and L_i^N and L_i^F are the inputs of native and foreign (immigrant) labor, respectively. Furthermore, $C^N(L_i^N)$ is the cost of native labor, and $C_i^F(L_i^F)$ is the potentially farm-specific cost associated with foreign labor. Note that in addition to the direct wage costs, these cost functions may include a fixed component, reflecting, for example, search and hiring costs, and they may also exhibit increasing marginal costs due to, for example, a limited local supply of labor.

If immigrants and natives are perfect substitutes and associated with the same cost, the profit function reduces to: $\pi_i = f_i(L_i) - C(L_i)$. In this case, immigrants simply represent a shift in the supply of labor with no farm-level relationship between the use of immigrants and performance. This corresponds to the first possibility mentioned in the introduction.

A causal effect of immigrants on farm performance may arise for at least two different reasons. First, immigrants may constitute a cheaper source of labor than natives for some farms, and these farms can then use

immigrant labor to reduce their total costs. This will be the case on farm i if $C^N(L_i^*) > C_i^F(L_i^{F*}) + C^N(L_i^* - L_i^{F*})$, where L_i^* is the optimal input of labor when only native workers are used, and where $L_i^{F*} \in (0, L_i^*]$ is chosen to minimize $C_i^F(L_i^F) + C^N(L_i^* - L_i^F)$. In this case, access to immigrant labor will increase profit on farm i , and possibly also total employment. The latter requires that the marginal cost of labor is also reduced, that is, that $C^{N'}(L_i^*) > C_i^{F'}(L_i^{F*})$. Native labor, on the other hand, may be negatively affected (crowded out) by the employment of immigrants.

Second, immigrants may constitute a more flexible type of labor that complements the skills of native employees. This requires a positive cross derivative in the production function of farm i . In this case, labor productivity, profits, and employment (total as well as native) may be positively affected by access to immigrant labor.

As access to (i.e., the cost of using) immigrant labor is likely to vary exogenously both across farms and over time, only some farms will use (or start using) immigrants. As an example, geographical differences as well as differences in language skills and attitudes towards immigrants may imply different hiring costs for different farms for reasons not related to farm performance. Furthermore, a limited and varying local supply of immigrant labor may imply that the marginal costs of using immigrants may also be higher at some farms than at others. In addition, the marginal costs may vary over time due to randomness and frictions in the search process, and because the general access to immigrant labor changes. As a result, we would only expect some farms to use (or to start/stop using) foreign labor, which creates an observed positive causal relationship between the use of immigrants and farm performance across farms and time.

An observed (positive) relationship between immigrants and performance may also arise for non-causal reasons. First, assume a fixed cost of hiring immigrants, for example, due to language barriers; thus, C_i^F contains a term, $c_0 \cdot I(L_i^F > 0)$, that is strictly positive but only included when $L_i^F > 0$. Furthermore, assume that the production function contains a farm-specific productivity (or scale) parameter, A_i , that impacts optimal farm size: $f_i(L_i^N, L_i^F) = f(A_i, L_i^N, L_i^F)$, where $f''_{12} > 0$ and $f''_{13} > 0$. In this formulation, farms with high values of A_i will be larger, all else being equal, and therefore also more likely to find it optimal

to incur the fixed cost of hiring immigrants. At the same time, these farms will also be more productive and more profitable. As a consequence, we will observe a positive correlation between the use of immigrants and farm performance, even if the only difference between immigrants and natives is that part of the cost of using immigrants is in the form of a fixed cost.

Second, assume that instead of containing a fixed element, the cost of using foreign labor, $C_i^F(L_i^F)$, is correlated with farm productivity, A_i . Then a non-causal relationship between the use of immigrants and farm performance may also emerge. This could be the case if, for example, certain types of farms or certain subsectors, such as nurseries, find it easier to use foreign labor, perhaps because tasks are more standardized here, and these farms/subsectors are at the same time the more (or less) productive/profitable ones.

As noted in the introduction, the different possibilities are, of course, not mutually exclusive. Thus, we could easily imagine a situation in which some of the less productive (or less advanced) farms have lower costs of using immigrants, but that the use of immigrants improves their performance compared to similar farms.⁴

Empirical Strategy

The theoretical model of the previous section implies that if immigrants have a causal effect on performance, we should expect farms with a larger share of immigrants to perform better. These farms have lower costs and/or are better able to exploit the complementarities between native and immigrant labor. Hence, the theoretical model suggests an empirical approach where farm performance is regressed on the share of immigrants in farm employment.

However, the theoretical section also stressed that unobserved farm-specific factors (A_i) may be correlated with both performance and the share of immigrants, thereby giving rise to a non-causal relationship between the share of immigrants and farm performance. In other words, there exists an omitted-variable

⁴ In principle, we can also imagine a negative causal effect of immigrants. If the costs of using native labor are also farm-specific (reflecting, e.g., local supply conditions), then some farms may be "forced" to use the less productive/more expensive immigrant labor.

bias in the regression of farm performance on the share of immigrants. To deal with this problem, we can use fixed effects (FE) and/or regressions in first differences (FD). We will rely on the latter approach, regressing changes in farm performance on changes in the share of immigrants. In both cases, however, the effect of immigrants will be identified from changes over time within farms. In other words, the idea behind our identification strategy is that changes over time in the exogenous farm-specific cost of using immigrants—possibly as a result of an improved (or more limited) access to immigrant labor—will lead some farms to use more (or fewer) immigrants, and this will in turn affect the performance or outcome of these farms.

To reduce measurement error and yearly noise, we use 4-year differences for the variables of interest. Furthermore, as some of the effects may take time to materialize, long differences may also better capture the true effects. Hence, our basic empirical specification looks like:

$$(2) \quad \Delta P_{it} = \beta_0 + \beta_1 \cdot \Delta Imm_{it} + \varepsilon_{it},$$

where ΔP_{it} is a measure of the change in performance (or outcome) of farm i between year t and year $t + 4$, ΔImm_{it} is the change in the share of immigrants in employment on farm i between t and $t + 4$, and ε_{it} is the error term. In some of the analyses, we also consider the change in performance between t and $t + 8$.

Estimating equation (2) may still suffer from an omitted-variable bias if unobserved (demand and supply) shocks affect both changes in farm performance and changes in the share of immigrants. An example could be an increase in the world-market price of pork, which increases farm profit and the demand for additional labor for some farms. If the marginal cost of hiring immigrants is lower than the marginal cost of hiring additional natives, for example, because it becomes worthwhile to incur the fixed cost of hiring immigrants, the result will be both an increase in performance and an increase in the use of immigrants for some farms.

To deal with this possibility, we first include dummy controls for each combination of sub-sector, region, and year. This is done to capture the effects of any region- and/or sub-sector-specific shocks to supply and demand (including any business-cycle effects). We also include a number of initial (i.e., time t) farm characteristics, such as age and size of the farm, as

well as the average experience and tenure of its employees, to provide a proxy for exposure and responsiveness to shocks. Third, we include farm-fixed effects in some regressions, which remove any farm-specific trends in performance from the data.

The resulting empirical specification then looks like:

$$(3) \quad \Delta P_{it} = \beta_0 + \beta_1 \cdot \Delta Imm_{it} + \beta_2 \cdot X_{it} + \eta_{rsy} + \mu_i + \varepsilon_{it},$$

where X_{it} is a vector of observable farm characteristics at time t , η_{rsy} represents the dummies for each combination of region (r), sub-sector (s) and year (y), and μ_i is an unobserved time-invariant, farm-specific trend in performance.

Still, there is a risk that unobserved non-permanent farm-specific shocks may affect both changes in farm performance and changes in the share of immigrants, and that these are not picked up by the included controls in equation (3). An example could be an invention or a change in management practices that causes a farm to switch crops, thereby changing both the performance and the demand for additional labor.

To deal with this possibility, we try to instrument the change in the share of immigrants between t and $t + 4$ using two types of instruments. First, we use historical shares of immigrants from different countries (or groups of countries) of origin (see next section) in farm employment to allocate subsequent inflows of immigrants at the aggregate level to the individual farms. More precisely, the instrument for ΔImm_{it} is constructed as:

$$(4) \quad \sum_{g=1}^G imm_{git-s} \cdot \Delta AggEmpl_{gt},$$

where imm_{git-s} is the share of immigrants from country (group) g employed on farm i at time $t - s$, and $\Delta AggEmpl_{gt}$ is the absolute nationwide or regional change in employment of immigrants from country (group) g in the agricultural sector between t and $t + 4$. The idea behind this instrument is that farms that have historically used immigrants in their workforce have lower marginal costs of employing additional immigrants of similar or related origin, and that the employment of additional immigrants from a given area is easier/cheaper in periods with a large inflow from that area at the aggregate level. Similar types of instruments

have previously been used by, for example, Card (2001) and Cortes (2008) to measure the regional shares of immigrants.⁵

Second, we simply use the lagged share of immigrants in farm employment, imm_{it-s} , as an instrument for ΔImm_{it} . This is a common strategy in models in first differences (Wooldridge 2002). The idea behind this instrument in the current context is a “reversion-towards-the-mean” argument. The agricultural sector (and the Danish economy more generally) is characterized by a high degree of job turnover. Hence, farms with an “above average” share of immigrants are likely to see their shares reduced (relative to other farms) in the future, simply because some of the immigrants quit (and possibly return home). If the cost of hiring new immigrants to replace the old ones varies over time for the individual farm, reflecting, for example, the current local supply and/or randomness/frictions in the search process, some of these farms may not, in the short run, be able to find a new immigrant worker to replace the one who left, which creates a negative correlation between the historical share and the subsequent change in the share of immigrants.

The instruments above are valid as long as we do not observe temporary shocks that affect both the share of immigrants s years in advance and the change in performance between t and $t + 4$, and which are not picked up by the included controls in equation (3). The larger we make s , the smaller this risk is likely to be.^{6,7}

As our dependent variable, we use a range of different measures of changes in performance: job creation and survival, growth in revenue, and growth in labor productivity. As we do not have any measures of capital and land at the farm level, we cannot construct a profit measure or a measure of total factor

productivity. To analyze whether immigrant labor complements or is a substitute for various types of native labor, we consider immigrant labor's effects on native job creation (both skilled and unskilled jobs), as well as its effects on the wages of individual native workers.

Data

We use data from the Integrated Database for Labor Market Research (IDA) compiled by Statistics Denmark. The IDA contains annual register data from 1980 to 2008 at the individual level regarding labor-market status and performance (employment, wages, etc.), and personal background characteristics such as age, education, and immigrant status. Wage and employment information in the IDA concerns primary employment in the last week of November each year. Furthermore, all wage workers are linked to an establishment in the IDA, and both individuals and establishments are tracked over time. From the IDA, we draw our “sample” containing all wage workers in a given year with primary employment in the agricultural sector, defined as establishments with NACE Rev. 2 codes between 01.11 and 01.50.⁸

Since individual workers are linked to establishments and because the data cover the population of workers and establishments, we can aggregate individual data at the establishment level and use this information about the establishment (e.g., the number of immigrants and natives employed, as well as the average wage, experience, tenure, and education of the employees) both at the establishment level and at the individual level. The establishment level information also includes the region and age of the farm, and a detailed industry classification that allows us to distinguish between four sub-sectors: *Arable farming*, *Livestock and mixed enterprises*, *Horticulture*, and *Other types of farming*.

Furthermore, as of 1998, establishments have been linked to firms (the legal unit). This provides us with information about revenue and value added (calculated as the difference between revenue and purchases of goods and services) at the firm level for the years 1998–2008. Note that the vast majority of firms consist of only one establishment.

⁵ In the empirical analyses, we use both individual origin countries and groups of origin countries in the construction of the instrument in equation (4), since an employer using immigrants from, for example, Poland may also see his/her marginal costs of using immigrants from neighboring countries reduced, as (s)he then knows how to recruit workers from that area and has changed the organization of the farm to accommodate these workers. We also use the individual terms in equation (4) to measure the changes in the shares of immigrants from different groups of origin countries.

⁶ Alternatively, we could use pre-sample shares of immigrants rather than shares in $t - s$ in the construction of the first instrument. This would increase the validity of the instrument. However, given the length of our sample period and the very limited number of immigrants in the beginning of the sample, such an instrument would become very weak.

⁷ Note that since the instruments are constructed from lagged values of the share of immigrants, we do not simultaneously include the farm-fixed effects, μ_i , in the IV regressions, as the demeaned error term in an FE regression would be correlated with lagged values of the regressors (e.g., Wooldridge 2002).

⁸ NACE is the statistical classification of economic activities used in the European Community. See http://epp.eurostat.ec.europa.eu/cache/ITY_OFFPUB/KS-RA-07-015/EN/KS-RA-07-015-EN.PDF.

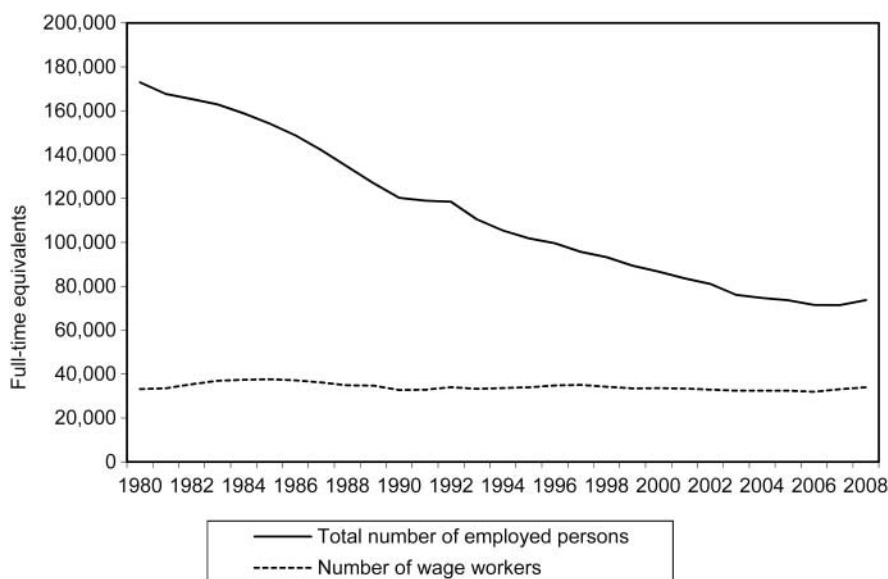


Figure 1. Total employment and wage workers in agricultural sector, 1980–2008

Note: The number of wage workers covers all persons (in full-time equivalents) with a primary occupation as wage worker in the agricultural sector in the last week of November each year. The total number of employed persons also includes self-employed. Wage workers are persons whose primary occupation is wage employment irrespective of their age.

Immigrants are defined as persons born outside Denmark to non-Danish parents, that is, to parents who do not have Danish citizenship or were themselves born outside of Denmark. If no information about the parents is available, an individual born outside Denmark is considered as an immigrant. Consequently, all individuals born in Denmark are considered to be native Danes, irrespective of the status of their parents, just as all individuals born abroad who have at least one Danish parent are considered as Danes. This definition also implies that immigrants include refugees and family reunified persons who have come to Denmark for non-job related reasons, as well as foreigners who have come to Denmark primarily to work. The definition of an immigrant's country of origin is based on the parents' countries of birth (or citizenship) whenever that information is available. That is, an immigrant born by Italian parents in Canada is considered Italian.

Traditionally, agriculture has been extremely important in Denmark, in terms of GDP, employment, and exports. For example, just after WWII, the agricultural sector was responsible for approximately 30% of the employment in Denmark (Statistics Denmark, 2001). Figure 1 shows the development in total employment as well as the number of wage workers in the Danish agricultural sector since 1980. The difference between the two

curves is the number of self-employed, which includes both self-employed farmers working alone and self-employed with employees, that is, those running a personally-owned establishment. From the figure, we can see that while the number of wage workers has been relatively constant around 30,000–40,000, the number of self-employed has been rapidly declining—from around 140,000 in 1980 to around 40,000 in 2008. This mirrors the trend towards fewer but larger farms as shown in figure 2.

Even though the definition of establishments excludes farms without employees (i.e., farms run by a single self-employed individual), figure 2 shows a sharp decline in the number of establishments, from around 20,000 in 1980 to below 11,000 in 2008. This also reveals that the majority of the 40,000 self-employed in 2008 do not have employees. Finally, figure 2 also shows a significant increase in the average size of an establishment from less than 2.4 employees (excluding the owner if the farm is personally-owned) in 1980 to 3.7 employees in 2008.

Figure 3 also shows a pronounced increase in the use of immigrant workers in the agricultural sector, especially since the mid-1990s. Until then, the agricultural sector relied significantly less on immigrant workers than the economy in general. The share among wage workers in 1980 was 1.6% and increased

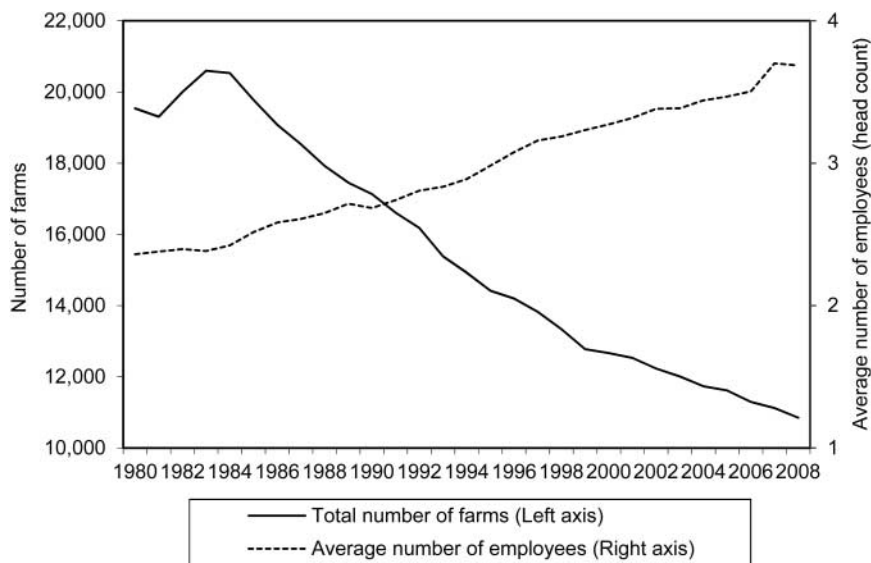


Figure 2. Number and average size of agricultural establishments, 1980–2008

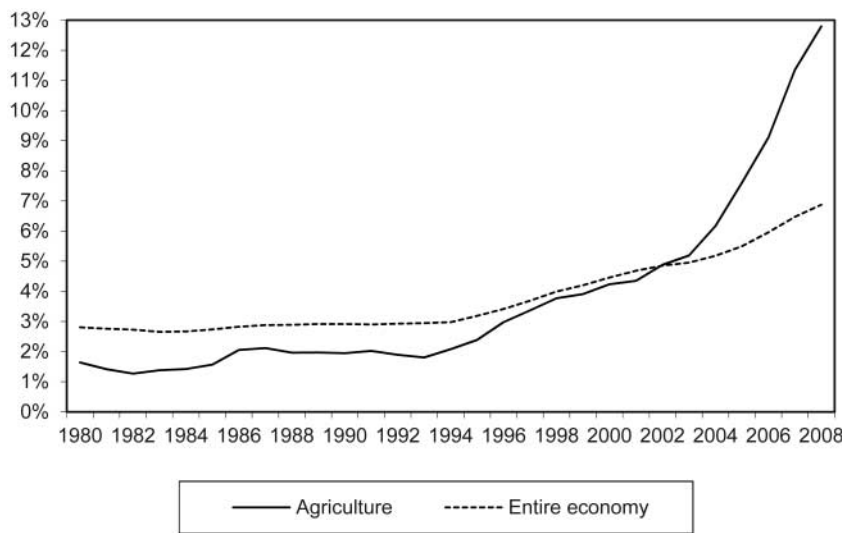


Figure 3. The share of immigrants among wage workers, 1980–2008

Note: The immigrant shares are calculated from the number of wage workers (in full-time equivalents) in agriculture and the entire economy, respectively, in the last week of November each year. Wage workers are persons whose primary occupation is wage employment irrespective of their age.

slightly to around 2% in 1993. After that, the use of immigrants in agriculture increased far more rapidly than in the rest of the economy. In 2008, immigrants thus made up 13% of the wage workers within agriculture, but only around 7% of the wage workers within the economy as a whole.

When analyzing the role of immigrants, in some of the analyses below we will distinguish between three groups of origin countries for the immigrants: first, immigrants from Western Europe, the United States, Canada,

Australia, New Zealand, and Japan (“Western immigrants”); second, immigrants from Eastern Europe, which includes Albania, Belarus, Bosnia and Herzegovina, Bulgaria, Croatia, Czech Republic, Estonia, Hungary, Latvia, Lithuania, Moldova, Montenegro, Poland, Republic of Macedonia, Romania, Russia, Serbia, Slovakia, and Ukraine; and third, immigrants from other (mainly less developed) countries, including Turkey and Pakistan, as well as Asian and African countries (“LDC immigrants”).

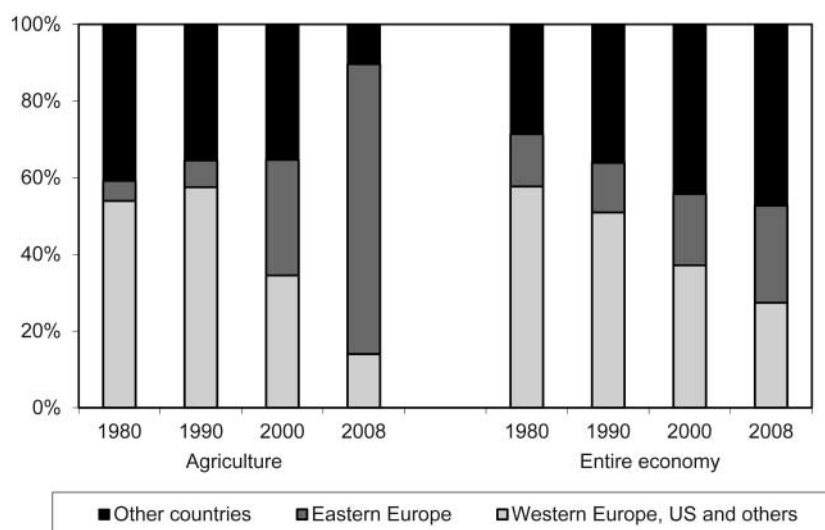


Figure 4. Distribution of immigrant wage workers according to countries of origin, 1980, 1990, 2000, and 2008

Note: Wage workers are persons whose primary occupation is wage employment irrespective of their age.

Due to an increased number of refugees and “family-reunified” persons, LDC immigrants have increased in relative importance over the last thirty years when the entire economy is considered; still, they constitute a small and decreasing share of immigrants within agriculture, as shown in figure 4. Indeed, the agricultural sector has seen a marked relative increase in the use of Eastern European immigrants from about 5% in 1980 to about 75% of the immigrants in 2008. The share of Western immigrants among immigrant wage workers has diminished in agriculture, and in the economy more generally, despite the fact that these immigrants have had more easy access to the Danish labor market than the other groups of immigrants, and the fact that the language barrier is likely to be smaller for this group than for the other two groups.

This development suggests that immigrants may have played a different role in agriculture than in the rest of the economy, as also suggested by Partridge, Rickman, and Ali (2008). The large and increased concentration of Eastern European immigrants in the agricultural sector (even before the enlargements of the European Union in 2004 and 2007, which granted free access to the Danish labor market for workers from 10 Eastern European countries⁹), combined with the fact that wages

in Eastern Europe are considerably lower than in Western Europe, also indicates that these immigrants may have played a different role than other immigrants in the agricultural sector.

Table 1 compares average key values of farms that employ immigrants to average key values of farms that only employ native workers for the years 1980, 1990, 2000, and 2008. Farms that use immigrant labor are significantly larger in terms of employees, although this difference has become smaller over time. From the firm-level data, we can also see that total revenue and value added were on average much higher on farms employing immigrants in both 2000 and 2008. Revenue per employee was also higher in 2008, while the lower value added per employee can almost completely be explained by the larger concentration of farms with immigrants within *Live-stock and mixed enterprises*, a sub-sector which in general experienced negative value added in 2008.¹⁰ If farms with more employees are also more labor-intensive (as opposed to land- or capital-intensive) than other farms, we should

in sectors that experienced bottlenecks, such as agriculture. Local occupation councils in which both employers and employees were represented decided whether the particular area had a bottleneck.

¹⁰ As most farms are likely to be owned and run by a self-employed farmer who is not included among the employees, revenue and value added per employee have been computed by increasing the number of employees by one to account for the labor input of the owner.

⁹ Before enlargement of the European Union in 2004, it was possible for Danish firms to recruit Eastern European workers

Table 1. Summary Statistics of Farm Characteristics, 1980, 1990, 2000, 2008

	Farms without immigrants								Farms with immigrants							
	1980		1990		2000		2008		1980		1990		2000		2008	
	# Obs	Mean	# Obs	Mean	# Obs	Mean	# Obs	Mean	# Obs	Mean	# Obs	Mean	# Obs	Mean	# Obs	Mean
Establishment-level data:																
<i>Number of employees</i>	19,085	2.094	16,541	2.400	11,338	2.643	7,821	2.544	454	13.551	592	10.657	1,326	8.662	3,027	6.138
<i>Arable farming</i>	19,085	0.843	16,541	0.840	11,338	0.188	8,265	0.278	454	0.513	592	0.576	1,326	0.106	2,856	0.115
<i>Livestock and mixed enterprises</i>	n.a.	n.a.	n.a.	n.a.	11,338	0.612	8,265	0.500	n.a.	n.a.	n.a.	n.a.	1,326	0.584	2,856	0.699
<i>Horticulture</i>	19,085	0.080	16,541	0.070	11,338	0.065	8,265	0.054	454	0.284	592	0.299	1,326	0.204	2,856	0.109
<i>Other types of farming</i>	19,085	0.077	16,541	0.090	11,338	0.135	8,265	0.168	454	0.203	592	0.125	1,326	0.106	2,856	0.077
<i>Share of workers with vocational education</i>	19,085	0.135	16,541	0.206	11,338	0.349	7,821	0.430	454	0.240	592	0.292	1,326	0.295	3,027	0.272
<i>Share of workers with higher education</i>	19,085	0.005	16,541	0.008	11,338	0.014	7,821	0.028	454	0.035	592	0.036	1,326	0.030	3,027	0.030
<i>Average tenure of employees (years)</i>	0	0	16,541	1.880	11,338	2.441	7,821	3.311	0	0	592	1.713	1,326	1.623	3,027	1.701
<i>Average experience of employees (years)</i>	19,085	4.515	16,541	6.194	11,338	8.387	7,821	11.921	454	4.662	592	5.836	1,326	5.381	3,027	5.774
<i>Average hourly wage (Immigrants), DKK</i>	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	155	67.730	337	103.992	591	128.859	1,817	163.006
<i>Average hourly wage (Danes), DKK</i>	4,294	60.462	10,266	98.550	8,643	133.012	6,645	180.220	271	65.407	411	106.312	821	139.992	1,952	183.439
<i>Average hourly wage (All persons), DKK</i>	4,294	60.462	10,266	98.550	8,643	133.012	6,645	180.220	288	65.541	487	104.554	956	135.150	2,510	173.687
Firm-level data:																
<i>Revenue, DKK 1,000</i>	n.a.	n.a.	n.a.	n.a.	10,857	3,796.20	7,506	5,254.4	n.a.	n.a.	n.a.	n.a.	1,257	6,460.07	2,948	13957.2
<i>Revenue/employees, DKK 1,000</i>	n.a.	n.a.	n.a.	n.a.	10,857	1,138.20	7,506	1,547.0	n.a.	n.a.	n.a.	n.a.	1,257	886.72	2,948	1,693.8
<i>Value added, DKK 1,000</i>	n.a.	n.a.	n.a.	n.a.	10,857	629.97	7,506	488.8	n.a.	n.a.	n.a.	n.a.	1,257	1,446.21	2,948	1,877.3
<i>Value added/employees, DKK 1,000</i>	n.a.	n.a.	n.a.	n.a.	10,857	164.13	7,506	101.8	n.a.	n.a.	n.a.	n.a.	1,257	143.92	2,948	3.2

Note: Summary statistics are computed as simple averages across all farms in the data set. Until after 1990, only three subsectors are used in the data, and the distribution across subsectors in 2008 is based on data from 2007. By construction, tenure equals zero in 1980. *Value added* is calculated as the difference between revenue and purchases of goods and services. *Revenue/employees* and *value added/employees* are constructed by adding 1 to the number of employees to account for the fact that most establishments are likely to be run by a self-employed owner who is not included among the employees.

actually expect revenue and value added per employee to be lower on larger farms, all else being equal. In light of this, and because farms with immigrants have on average more than twice as many employees, the numbers for revenue and value added per employee seem to indicate that farms using immigrants are, on average, at least not less productive than farms not employing immigrants.¹¹

This conclusion is further supported by the fact that throughout the period average wages of native workers have been higher on farms that employ immigrants, although less so in later years. Wages to immigrants, on the other hand, are significantly lower, except in 1980, and this difference has increased over time. This is a first indication that immigrants constitute a cheaper source of labor, although if we control for differences in individual worker productivity, the effective wage of immigrants need not be lower.

Table 1 also reveals that throughout the period, farms with immigrants were more prevalent within *Horticulture*. Furthermore, in 1980 and 1990, the shares of employees with either vocational or higher education (“skilled workers”) were also higher on farms that used immigrants than on other farms. However, this picture was reversed by 2008. Conversely, while there were no clear differences in average tenure and average experience between farms with and without immigrants until 1990, farms employing immigrants exhibited considerably less average experience and average tenure among their employees in 2000 and 2008.

Table 2 compares the wages and educational levels of natives and the three different groups of immigrants considered in the article. Note, however, that information on education and wages is missing for a large number of the immigrants. Immigrants with missing information about education are treated as unskilled in table 2. Hence, the numbers should be interpreted with some caution. Still, among the immigrants, wages have been lowest for immigrants from Eastern Europe since the 1990s. In 2008, these immigrants’ average hourly wage was 20% lower than that of natives, while Western immigrants had a wage level much closer to the wages of natives. Back in 1980, only LDC immigrants had lower wages than native workers. Immigrants from LDCs and, especially, Eastern Europe have also tended to

be less educated than natives in more recent years.¹²

In summary, farms that employ immigrant workers both are larger in terms of employees, revenue, and value added and seem to be at least as productive as farms without immigrants. This is consistent both with a selection effect in the form of a fixed cost of employing immigrants (or even a negative correlation between this cost and farm scale/productivity), and with a positive causal effect of immigrants. However, while the descriptive evidence reveals clear differences between the two types of farms, and between immigrants and natives, it cannot disentangle any causal effects from selection effects.

Immigrant Workers and Farm Performance

In this section, we analyze the relationship between the employment of immigrants and farm performance in more detail by following the empirical strategy laid out above. First, we use our establishment-level data, which cover the entire period of 1980–2008, to analyze the effects on short- and long-term job creation, as well as establishment survival. We then turn to our firm-level data, which cover the period 1998–2008, to analyze the effects on revenue and labor productivity (measured as revenue and value added per employee). Finally, we consider the effects of employing immigrants on native workers to determine how immigrants complement or substitute for different types of native labor.

As the variables of interest in most regressions are defined as differences between t and $t + 4$, and as we need lagged variables to construct the instruments, we use $t = 1984, 1988, 1992, 1996, 2000$, and 2004 in the establishment-level data, and $t = 2000$ and 2004 in the firm-level data. In the long-term job-creation and survival analyses, the dependent variable is a difference between t and $t + 8$, in which case we need to drop $t = 2004$ as well.

Job Creation and Establishment Survival

We start by analyzing the effects of employing immigrants on total job creation at the establishment level. We follow Davis and

¹¹ Unfortunately, we do not have information on capital and land holdings that would allow us to construct a measure of profitability or total factor productivity at the farm level.

¹² Another dimension of the importance of immigrants in the agricultural sector is their role as employers; immigrants’ share among self-employed owners is still limited, but rose from 2.1% in 2004 to 2.6% in 2008.

Table 2. Wages and Educational Attainment of Immigrant and Native Workers

	1980		1990		2000		2008	
	# Obs	Mean	# Obs	Mean	# Obs	Mean	# Obs	Mean
Immigrants from Western Europe, USA, etc. (group 1)								
Share of workers with vocational education	359	0.365	404	0.376	535	0.484	646	0.430
Share of workers with higher education	359	0.089	404	0.064	535	0.088	646	0.094
Average hourly wage, DKK	121	69.04	227	107.05	342	138.97	495	182.79
Immigrants from Eastern Europe (group 2)								
Share of workers with vocational education	99	0.273	142	0.317	1,074	0.137	5,223	0.070
Share of workers with higher education	99	0.081	142	0.085	1,074	0.034	5,223	0.019
Average hourly wage, DKK	29	72.07	49	107.63	235	122.77	2,207	157.36
Immigrants from other (LDC) countries (group 3)								
Share of workers with vocational education	251	0.143	282	0.202	664	0.202	525	0.271
Share of workers with higher education	251	0.020	282	0.050	664	0.036	525	0.074
Average hourly wage, DKK	62	61.74	174	99.38	396	125.98	391	171.06
Native workers								
Share of workers with vocational education	45,404	0.176	45,175	0.263	39,174	0.400	33,594	0.478
Share of workers with higher education	45,405	0.019	45,175	0.021	39,174	0.029	33,594	0.043
Average hourly wage, DKK	13,785	64.50	27,261	108.98	27,899	145.45	27,702	194.96

Note: Summary statistics are computed as simple averages across all individuals in the data set.

Haltiwanger (1992) and Davis, Haltiwanger, and Schuh (1996) and define the net job-creation rate between t and $t + 4$ at establishment i as:

$$(5) \quad \Delta x_{it+4} = (X_{it+4} - X_{it}) / (0.5(X_{it+4} + X_{it})),$$

where X_{it} is the employment at establishment i in year t . An advantage of using this measure instead of a standard growth rate is that the range of Δx_{it+4} is bounded between -2 and 2 and hence prevents outliers. This is particularly relevant for small establishments, where changes in employment can be large relative to their initial levels.

Table 3 contains the results of estimating the empirical models in equations (2) and (3) using the net job-creation rate from equation (5) as the performance measure, ΔP_{it} .

In columns 1 and 2, we present the results of estimating the stripped-down version of our empirical model from equation (2) using OLS, where the only explanatory variable used is the change in the share of immigrants at establishment i between t and $t + 4$. In the first

column, we use the overall share of immigrants in employment, and in column 2, we distinguish between our three country groups of immigrants. There is a significant positive relationship between the change in the overall share and the net job-creation rate in column 1. This relationship is found in column 2 to be driven by immigrants from Eastern Europe and LDCs, while there is no relationship between the change in the share of Western immigrants and the net job creation rate.

As argued previously, the estimates in columns 1 and 2 may suffer from an omitted-variable bias if unobserved shocks affect both the use of immigrants and overall job creation. In columns 3 and 4, we therefore add dummies for each combination of year, sub-sector (four categories), and region (three categories) to capture any effects of region- and subsector-specific shocks. We also add the following farm characteristics at time t : age of the establishment (25 dummies); number of employees (11 dummies); average tenure of the employees; and average experience of the employees. This hardly affects the coefficient estimates to Δimm_{it} . Furthermore, adding farm fixed

Table 3. Employment of Immigrant Workers and Job Creation

	Dependent variable: Job creation rate between t and $t + 4$								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	FD	FD	FD	FD	FD + FE	FD + FE	FD + IV	FD + IV	FD + IV
<i>Change in total share of immigrants in farm employment between t and $t + 4$</i>	0.1132 (7.02)***		0.1054 (7.56)***		0.0826 (5.90)***		0.2193 (1.22)	0.2192 (1.22)	
<i>Change in share of immigrants between t and $t + 4$ from:</i>									
<i>Western Europe, USA, etc. (group 1)</i>		0.0064 (0.25)		0.0125 (0.59)		0.0146 (0.74)			−0.0788 (−0.37)
<i>Eastern Europe (group 2)</i>		0.1582 (7.78)***		0.1463 (8.05)***		0.1128 (5.93)***			0.5396 (1.24)
<i>Other (LDC) countries (group 3)</i>		0.1439 (2.96)***		0.1386 (3.34)***		0.1230 (2.90)***			1.2096 (3.10)***
Region $\times$ sub-sector $\times$ year dummies	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Farm controls	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Farm fixed effects	No	No	No	No	Yes	Yes	No	No	No
Observations	54,622	54,622	54,622	54,622	54,622	54,622	45,265	45,265	45,265
Number of establishments	22,041	22,041	22,041	22,041	22,041	22,041	18,074	18,074	18,074
R-squared	0.0015	0.0021	0.1905	0.1910	0.4906	0.4909	0.1799	0.1799	0.1549

Note: Estimations are based on a panel of farm establishments from 1980 to 2008. Results above are for $t = 1984, 1988, 1992, 1996, 2000$, and 2004 . Region $\times$ subsector $\times$ year dummies are constructed from three regions and four subsectors. The farm controls used are age of establishment at time t (25 categories), number of employees at time t (11 categories), average tenure at time t , and average experience at time t . IV estimations are implemented as 2SLS. In column 7, both the lagged share of immigrants from $t - 2$ and the inflow-augmented lagged share from $t - 2$ are used as instruments. In columns 8 and 9 only the lagged share(s) from $t - 2$ is used. First stage F-statistics are as follows: 25.0 (column 7), 50.0 (column 8) and 18.2, 4.1, and 9.8 (column 9). The R-squared reported in columns 5 and 6 is the “within R-squared,” i.e., from a regression on the demeaned data. t -values in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. Standard errors are clustered at the establishment level.

effects (FE) in columns 5 and 6 to control for farm-specific trends does little to change this picture, although the estimates become slightly smaller numerically.

Finally, in columns 7–9, we implement the IV strategy described above using two-stage least squares (2SLS). In column 7, we use both of the proposed instruments with $s = 2$; that is, we use the share of immigrants in $t - 2$ to predict the change in the share of immigrants between t and $t + 4$.

Unfortunately, our first instrument from equation (4) was found not to work. Although it had the expected sign in the first stage, the coefficient estimate was far from significant. We tried using different country groupings, different lags, and nationwide or regional inflows of immigrants into the agricultural sector, but without success. For the results reported in column 7, 33 country groups (the 10 most important countries, plus a residual group within each of the three overall country groups) and nationwide inflows from these groups were used to construct the instrument.

A possible explanation for the weakness of the first instrument may be that relatively few farms have more than one immigrant among their employees (approximately 3.1% in 2000, and 13.3% in 2008, corresponding to 30% and 48%, respectively, of the farms with immigrants), and among these only around half also had immigrants employed six years earlier. This makes it difficult to predict increases in the share of immigrants based on historical shares of immigrants at the farm level.¹³ In other words, most farms use very few immigrants (typically just one), and a large part of the increase in immigrant employment has thus taken place at the extensive margin, that is, due to an increase in the number of farms using immigrants. Furthermore, job spells in the agricultural sector are relatively short, especially on farms using immigrants, as evidenced by the tenure values in table 1.

This may also explain why the second instrument works much better with a negative and significant coefficient in the first stage. Very similar and more significant first-stage results were obtained by using the share of immigrants in $t - 1$ (i.e., $s = 1$), but as argued above, a smaller value of s also increases the risk of invalidating the instrument. We therefore also tried using longer lags (which made the

instrument weaker) and using several lags, but with similar results. Hence, in columns 8 and 9 and in the remainder of this article, we use only the second instrument with $s = 2$.

In column 8, the F-statistic from the first stage is 50, which is well above the critical value of 10 suggested by Staiger and Stock (1997) and Stock and Yogo (2002) to avoid problems with weak instruments, while the three F-statistics corresponding to the three first-stage regressions in column 9 vary between 4.1 and 18.2.

While the coefficients to the overall share of immigrants in columns 7 and 8 are still positive and somewhat larger than before (0.22), the standard errors have also increased considerably, resulting in insignificant estimates in both columns (the t -values are 1.22). But when we distinguish between the three groups of immigrants (column 9), using the shares of each of these in $t - 2$ as instruments, we find a significantly positive effect of immigrants from LDCs. (The effect is also positive but insignificant in the case of Eastern European immigrants.)¹⁴

Note that the IV estimate is numerically much larger than the non-IV estimate(s), but it also has a considerably higher standard error. Hence, we choose not to place too much emphasis on the precise value of the estimate. Instead, we interpret it as evidence supporting the idea that the positive effects of employing (LDC and Eastern European) immigrants found in columns 1–6 at least partly reflect a causal effect of these on employment growth at the farm level, and that the effect may be quantitatively more important than suggested by the OLS estimates in columns 1–6.¹⁵ For the same reason, we do not

¹⁴ This effect was confirmed in a 2SLS regression (not reported) where we only included the change in the share of LDC immigrants (and used its own lagged share as the instrument). Here the point estimate was slightly smaller and equally significant, with a first-stage F-statistic around 23.

¹⁵ A quantitatively larger estimate in IV regressions is not uncommon (e.g., Angrist and Pischke 2009). As an example, Angrist and Kugler (2003) find a similarly large difference between their OLS and IV estimates in an analysis of the employment effects of immigration. Another explanation for the quantitative difference between the OLS and IV estimates may be that the IV estimation retains only the variation in Δimm_{it} generated by the instrument, imm_{it-2} , and hence spans a different type of variation than Δimm_{it} . As the instrument captures that farms with an initially high share of immigrants experience a reduction in their share of immigrants (a negative Δimm_{it}) compared to other farms, the IV estimates reflect the effect of this on job creation, whereas the OLS estimate also captures the effects of increases in the share of immigrants. If the negative employment effect of losing an immigrant worker is quantitatively larger than the positive employment effect of hiring an immigrant worker, this may explain part of the difference between the OLS and IV estimates.

¹³ Another explanation may be that we eliminate much of the variation by including region $\times$ sub-sector $\times$ year fixed effects in the regressions.

necessarily interpret the difference in the IV results between LDC immigrants and immigrants from Eastern Europe as reflecting a difference in effects between the two types. The fact that LDC immigrants have been more constantly present throughout the considered period is likely to explain why the instrument works better in the case of these immigrants as indicated by the first-stage F-statistics, and hence why the IV estimate is only significant in this case. Instead, we are led to conclude that using both LDC immigrants and Eastern European immigrants seems to have a positive causal effect on job creation.¹⁶ Finally, the IV estimations confirm that using Western immigrants (where the instrument is strongest in the first stage with a value of 18.2 for the F-statistic) have no impact on the job creation rate. The point estimate is negative and insignificant.

Quantitatively, the estimates in columns 1–6 imply that an increase in the share of (Eastern European and/or LDC) immigrants of 10 percentage points is, on average, associated with a 1.5% increase in farm employment measured relative to the average employment in t and $t + 4$. Hence, the effects are relatively modest, and this indicates that the overall increase in employment takes place largely at the expense of native employment. We will return to this issue below. Even if we take the IV estimate from column 9 at face value, the absolute effect is not numerically very large: 11–12% from a change of 10 percentage points in the share of LDC immigrants, which is just high enough to avoid an adverse effect on native employment.

Thus, overall, the results in table 3 seem to support the idea that employing immigrants may have a positive causal effect on farm performance as measured by employment growth. In particular, positive effects seem to be associated with the use of immigrants from LDCs and Eastern Europe. As these were also the low-wage immigrants (table 2), this supports the idea that these immigrants constitute a cheap source of labor that reduces the marginal and average cost of labor for farms with access to these types of immigrants, and hence causes them to expand. That the increase in employment seems to take place largely, or at least partly, at the expense of native labor (unless

we take the IV estimate for LDC immigrants at face value) further indicates that these immigrants substitute for, rather than complement, native labor. We return to a more detailed analysis of this below. Immigrants from Western Europe, on the other hand, receive wages that are much more comparable to those of natives (table 2), and hence do not reduce the marginal cost of labor. Therefore, we do not observe any positive effects on job creation in this case.

To investigate whether there are also more “long-term” effects of employing immigrants, table 4 considers the effects on job creation between t and $t + 8$ from a change in the share of immigrants between t and $t + 4$. Otherwise, the empirical model and the explanatory variables are exactly as they are in table 3. The columns in table 4 also correspond to the columns in table 3, with the exception that we omit the column corresponding to column 7 from table 3.

It can easily be seen that the positive effects on job creation between t and $t + 4$ have disappeared by $t + 8$. Actually, with FE included (columns 5 and 6), point estimates tend to become negative but are only marginally significant in one case. The closest to a significantly positive estimate is the IV estimate for LDC immigrants in column 8 with a t -value of 1.58. Hence, it appears that the effect of immigrants on farm performance is rather short-lived. This fits well with the observation that average tenure is quite low. Thus, most of the immigrants hired between t and $t + 4$ will have left the farm by $t + 8$, thereby removing the cost advantage.

We also estimated the model using establishment survival to $t + 8$ (defined as a positive number of employees in $t + 8$) as the dependent variable.¹⁷ Establishment survival vs. closure is a clearer manifestation of changes in farm performance than (small) changes in employment, but the results (not reported) were very similar.

In summary, the positive (but small) short-run effects on employment of hiring immigrants from LDCs and Eastern Europe disappear relatively quickly. This is fully consistent with the idea these immigrants provide a cost advantage, but that access to them is only temporary.

¹⁶ We also estimated the regressions in table 3 separately for each of the four sub-sectors. While the results in columns 1–6 were rather robust across sectors, significant IV results were only found within *Horticulture*, which is not surprising given that the vast majority of LDC immigrants have been concentrated in *Horticulture*. However, the overall share of immigrants was also found to have a significant impact in this case.

¹⁷ Note that the closure of an establishment does not necessarily imply farm closure, as the farm may, in principle, continue without workers being employed. Unfortunately, our data do not allow us to control for this possibility.

Table 4. Employment of Immigrant Workers and Long-Run Job Creation

	Dependent variable: job-creation rate between t and $t + 8$							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	FD	FD	FD	FD	FD + FE	FD + FE	FD + IV	FD + IV
<i>Change in total share of immigrants in farm employment between t and $t + 4$</i>	0.0180 (0.51)		0.0037 (0.10)		−0.0617 (−1.80)*		0.0878 (0.15)	
<i>Change in share of immigrants between t and $t + 4$ from:</i>								
<i>Western Europe, USA, etc. (group 1)</i>		−0.0087 (−0.17)		−0.0160 (−0.30)		−0.0600 (−1.20)		−0.3141 (−0.48)
<i>Eastern Europe (group 2)</i>		0.0357 (0.70)		0.0141 (0.27)		−0.0694 (−1.36)		−0.7579 (−0.32)
<i>Other (LDC) countries (group 3)</i>		0.0363 (0.42)		0.0254 (0.30)		−0.0467 (−0.53)		2.6009 (1.58)
Region $\times$ sub-sector $\times$ year dummies	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Farm controls	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Farm fixed effects	No	No	No	No	Yes	Yes	No	No
Observations	47,525	47,525	47,525	47,525	47,525	47,525	39,287	39,287
Number of establishments	20,221	20,221	20,221	20,221	20,221	20,221	16,531	16,531
R-squared	0.0000	0.0000	0.0146	0.0146	0.3607	0.3607	0.0133	−0.015

Note: Estimations are based on a panel of farm establishments from 1980–2008. Results above are for $t = 1984, 1988, 1992, 1996, 2000$ and 2004 . Region $\times$ sub-sector $\times$ year dummies are constructed from three regions and four sub-sectors. The farm controls used are: Age of establishment at time t (25 categories), number of employees at time t (11 categories), average tenure at time t , and average experience at time t . IV estimations are implemented as 2SLS. In columns 7 and 8, the lagged share(s) of immigrants from $t - 2$ is used as instrument(s). 1st stage F-statistics are as follows: 25.2 (column 7) and 13.1, 2.5 and 8.4 (column 8). The R-squared reported in columns 5 and 6 is the “within R-squared”, i.e., from a regression on the demeaned data. t -values in parentheses. ***, ** and * indicate significance at the 1, 5 and 10% level, respectively. Standard errors are clustered at the establishment level.

Revenue and Labor Productivity

Job creation and farm survival are not the only relevant dimensions of farm performance to consider. Ideally, we would like to use a profit measure and/or a measure of total factor productivity on the individual farm. Unfortunately, information on profit and capital is not available in the data. But for the period 1998–2008, we do have measures of revenue and value added. Hence, in the following discussion we use the change in revenue as an alternative measure of farm growth, and revenue and value added per employee as measures of labor productivity.

Table 5 presents the results of estimating our empirical model in (3) on these data. The first four columns use the change in the log of total revenue between t and $t + 4$ as the dependent variable, whereas the next four columns use the change in the log of revenue per employee, and the last four columns use the change in value added per employee. Due to many negative observations of value added, we do not use logs in the latter case.¹⁸

Columns 1 and 2 present the results with region $\times$ sub-sector $\times$ year dummies and farm controls included, while columns 3 and 4 contain IV estimates.¹⁹ An increase in the share of immigrants is associated with an increase in revenue (column 1). Not surprisingly, the effect can be ascribed to immigrants from Eastern Europe (column 2), who in the considered period increased from approximately 1,000 to 5,000 in total employment.

Quantitatively, revenue increases by 0.8% if the share of Eastern European immigrants in employment increases from, for example, 0 to 0.1. This effect is similar to (although smaller than) the one found for job creation in table 3, and thus supports the interpretation of the results from the previous sub-section.

The IV results in columns 3 and 4 cannot confirm that this was due to the causal effect of immigrants. Note, however, that the first-stage F-statistics are also considerably lower than in the establishment-level data in table 3, indicating that the instruments work less well in this

case. This can be explained by the shorter time period considered.²⁰

The following four columns consider the effects on revenue per employee as a first crude measure of productivity, while the last four columns use the change in value added per employee. Most estimates in columns 5–12 are negative but insignificant. Only the IV estimate in column 11 is significantly negative at a 10% level. Hence, there are no signs that immigrants improve farm productivity, as we would expect if immigrants were complementing native labor. These results therefore also support the interpretation that (some) immigrants constitute a cheaper substitute for native labor. In this case, we would expect farm profitability and size to increase, but with no effects on labor productivity.

There may be at least three explanations behind the fact that the point estimates in columns 5–12 tend to be negative. First, given the importance of land in agriculture—a factor which is often in fixed supply—we would expect most farms to exhibit decreasing returns to scale and, hence, both measures of labor productivity to vary negatively with the amount of labor used (and thereby with increases in the share of immigrants according to the results from table 3). Second, the possibility that the effective labor input of an immigrant worker is lower than that of a native worker would also imply that the measures of labor productivity should decrease with increases in the share of immigrants, all else being equal. Third, employing cheap immigrant labor may lower the incentives to invest in capital, which in turn will show up as lower labor productivity.

In summary, the firm-level evidence reveals a small positive (possibly causal) effect on revenue, but no effects on labor productivity as measured by revenue and value added per worker. Both findings support the interpretation of the results from the job-creation analysis above, namely that especially LDC and Eastern European immigrants are used as a cheap substitute for native labor.

Native Employment

We found positive (short-run) effects of the employment of LDC and Eastern European

¹⁸ Furthermore, due to the presence of many (extreme) outliers in the data, we omit observations within the highest and lowest percentiles of one of the following variables: *Revenue*, *Value added*, *Revenue/employees*, and *Value added/employees* (measured in levels and in four-year changes).

¹⁹ Estimates with farm-fixed effects included are not reported. With only two periods, 2000–2004 and 2004–2008 in the data, including these effectively reduces the number of observations used by more than 50%.

²⁰ Also recall that the instrument is likely only to capture why farms with an initially high share of immigrants experience a reduction in their share compared to other farms. Hence, the instrument is unlikely to capture the main variation from the OLS estimation in columns 1 and 2, which is due to the large influx of Eastern European immigrants in recent years.

Table 5. Employment of Immigrant Workers, Revenue, and Labor Productivity

	LHS: change in log(revenue) from t to $t + 4$				LHS: change in log(revenue/ employees) from t to $t + 4$				LHS: change in value added/ employees from t to $t + 4$			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	FD	FD	FD + IV	FD + IV	FD	FD	FD + IV	FD + IV	FD	FD	FD + IV	FD + IV
<i>Change in total share of immigrants in farm employment between t and $t + 4$</i>	0.0696 (4.52)***		0.1258 (0.56)		-0.0242 (-1.40)		-0.0621 (-0.26)		-20.869 (-1.01)		-478.105 (-1.70)*	
<i>Change in share of immigrants between t and $t + 4$ from:</i>												
<i>Western Europe, USA, etc. (group 1)</i>		0.0290 (0.99)		-0.1357 (-0.46)		-0.0070 (-0.21)		-0.1143 (-0.37)		-26.171 (-0.56)		-495.673 (-1.35)
<i>Eastern Europe (group 2)</i>		0.0813 (4.71)***		0.4093 (1.11)		-0.0253 (-1.29)		0.1460 (0.42)		-23.928 (-1.00)		-518.480 (-1.21)
<i>Other (LDC) countries (group 3)</i>		0.0506 (0.87)		0.4499 (1.33)		-0.0500 (-0.76)		-0.6392 (-1.50)		17.165 (0.26)		-204.388 (-0.54)
Region $\times$ sub-sector $\times$ year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	13,277	13,277	10,451	10,451	13,277	13,277	10,451	10,451	13,289	13,289	10,458	10,458
Number of firms	9,141	9,141	7,328	7,328	9,141	9,141	7,328	7,328	9,150	9,150	7,332	7,332
R-squared	0.1037	0.1038	0.0785	0.0421	0.1416	0.1416	0.1509	0.1370	0.0145	0.0145	-0.0269	-0.0306

Note: Estimations are based on a panel of firms from 1998 to 2008. Results above are for $t = 2000$ and 2004. Revenue/employees and value added/employees are constructed by adding 1 to the number of employees to account for the labor input of a self-employed owner. Region $\times$ sub-sector $\times$ year dummies are constructed from three regions and four sub-sectors. The firm controls used are: Age of firm at time t (16 categories), number of employees at time t (11 categories), average tenure at time t , and average experience at time t . IV estimations are implemented as 2SLS. In columns 3, 4, 7, 8, 11, and 12, the lagged share(s) of immigrants from $t - 2$ is used as instrument(s). 1st stage F-statistics are as follows: 22.0 (columns 3, 7, and 11) and 9.7, 3.5 and 4.8 (columns 4, 8 and 12). t -values in parentheses. ***, ** and * indicate significance at the 1, 5, and 10% level, respectively. Standard errors are clustered at the firm level.

immigrants on job creation and, to some extent, revenue growth. The fact that these types of immigrants are also those with the lowest wages compared to natives, and that the employment growth took place largely at the expense of native labor, led us to conclude that these immigrants serve as a cheaper substitute for native labor, which allows farms to increase their scale of production. This interpretation is also consistent with the absence of any effects on labor productivity.

Despite this general picture, it could well be the case that different types of native workers are affected differently. In this section, we therefore analyze how the employment and wages of skilled and unskilled native workers, respectively, are affected by the employment of immigrants on the farm.

Table 6 considers the employment effects on natives. Columns 1–4 contain the effects on total native job creation (defined as the change in native employment between t and $t + 4$ relative to the average native employment in the two years) using exactly the same approach as in table 3. These results confirm the findings from table 3 (columns 1–6). Consistent with small or no effects on total job creation, we find large negative effects on native job creation in columns 1–2 of all three types of immigrants.²¹ That the effect is strongest for Western immigrants is fully in accordance with the finding from table 3 that there are no positive overall employment effects in this case.

The negative employment effect of the total immigrant share is confirmed in the IV regression in column 3. That the IV estimate is only significant for Western immigrants in column 4 can be explained in part by the stronger first stage in this case, but it is also consistent with the IV result from table 3, where the positive effect of LDC immigrants on total job creation was strong enough to eliminate the adverse effect on native employment. The IV estimates for Eastern European immigrants are still plagued by a weak first stage.

Columns 5–8 and 9–12 examine the effects on skilled and unskilled native job creation, respectively, to investigate which natives are replaced by immigrants. It turns out that there are significant negative effects on both types of natives in the OLS regressions (columns 5–6

and 9–10), but the effects are much stronger for unskilled natives. This is supported by the IV regressions. Here, the effect on skilled labor becomes insignificant, while it remains significantly negative in the case of unskilled labor, except for the case of LDC immigrants in column 12. In summary, immigrants seem (mostly) to replace unskilled labor with more limited negative effects on skilled native employment.

To analyze the wage effects on native co-workers who remain employed on-farm, we construct a data set at the individual level and run the regression in equation (3), where the left-hand side variable is replaced by the change in the (log) wage of a native co-worker on farm i between t and $t + 4$, given that this worker was employed on the farm in both t and $t + 4$. The results of this exercise are presented in table 7. Columns 1–4 consider the effects on skilled native co-workers, and columns 5–8 deal with unskilled native co-workers.

While there are no significant effects on the wages of unskilled native labor, there seems to be a positive effect on the wages of skilled native co-workers from an increase in the share of LDC immigrants (column 2). The effect, however, is small (a 10 percentage point increase in the share of LDC immigrants in employment is associated with an increase in the wage of 0.9%), and cannot be confirmed in the IV regressions (columns 3–4), possibly because the first stage is very weak. Still, it may be a weak indication that the skilled native workers who stay at the farm receive a stake in the cost advantage obtained by substituting low-skilled native labor with immigrants, perhaps because the skilled native workers specialize in more complex (managerial) tasks when immigrants are employed as suggested by, for example, Ottaviano, Peri, and Wright (2012).

In summary, the results of this section show that immigrants tend mostly to replace unskilled native workers. The effects on skilled native employment are less negative or at best absent. Furthermore, skilled native workers who remain employed on-farm seem to receive a (small) share in the cost advantage.

Conclusions

Many developed countries have experienced a marked increase in the use of immigrant workers in the agricultural sector. In Denmark, the share of immigrants rose from 2% in 1993 to

²¹ A 10 percentage point increase in the share of immigrants is, on average, associated with a decrease of 15% in native employment. Results with farm FE included are very similar to those in columns 1–2. This is also the case for the FE results corresponding to columns 5–6 and 9–10.

Table 6. Employment of Immigrant Workers and Native Job Creation

	LHS: native job-creation rate from t to $t + 4$				LHS: skilled native job-creation rate from t to $t + 4$				LHS: unskilled native job-creation rate from t to $t + 4$			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	FD	FD	FD + IV	FD + IV	FD	FD	FD + IV	FD + IV	FD	FD	FD + IV	FD + IV
<i>Change in total share of immigrants in farm employment between t and $t + 4$</i>	−1.6113 (−104.27)***		−1.0441 (−4.05)***		−0.7494 (−27.44)***		−0.0608 (−0.17)		−1.3067 (−53.70)***		−1.5227 (−4.53)***	
<i>Change in share of immigrants between t and $t + 4$ from:</i>												
<i>Western Europe, USA, etc. (group 1)</i>		−1.6376 (−62.72)***		−1.3561 (−4.72)***		−0.7020 (−15.22)***		−0.2907 (−0.65)		−1.3222 (−32.48)***		−1.5729 (−3.96)***
<i>Eastern Europe (group 2)</i>		−1.6314 (−80.46)***		−0.6346 (−0.90)		−0.8216 (−23.82)***		1.0116 (1.13)		−1.3064 (−41.99)***		−2.1536 (−3.01)***
<i>Other (LDC) countries (group 3)</i>		−1.4519 (−31.38)***		−0.0930 (−0.17)		−0.5229 (−7.85)***		−0.2528 (−0.38)		−1.2696 (−20.04)***		−0.5625 (−0.99)
Region × sub-sector × year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Farm controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Farm fixed effects	No	No	No	No	No	No	No	No	No	No	No	No
Observations	54,622	54,622	45,265	45,265	54,622	54,622	45,265	45,265	54,622	54,622	45,265	45,265
Number of establishments	22,041	22,041	18,074	18,074	22,041	22,041	18,074	18,074	22,041	22,041	18,074	18,074
R-squared	0.3559	0.3562	0.3188	0.2811	0.0395	0.0398	0.0250	−0.0191	0.1339	0.1340	0.1288	0.1133

Note: Estimations are based on a panel of farm establishments from 1980 to 2008. Results above are for $t = 1984, 1988, 1992, 1996, 2000,$ and 2004 . Region × sub-sector × year dummies are constructed from three regions and four sub-sectors. The farm controls used are: Age of establishment at time t (25 categories), number of employees at time t (11 categories), average tenure at time t , and average experience at time t . IV estimations are implemented as 2SLS using the lagged share(s) of immigrants from $t - 2$ as instrument(s). 1st stage F-statistics are as follows: 50.0 (columns 3, 7, and 11) and 18.2, 4.1, and 9.8 (columns 4, 8, and 12). t -values in parentheses. ***, **, and * indicate significance at the 1, 5, and 10% levels, respectively. Standard errors are clustered at the establishment level.

Table 7. Employment of Immigrant Workers and Native Wages

	LHS: change in log(wage) from t to $t + 4$							
	Skilled native workers				Unskilled native workers			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	FD	FD	FD + IV	FD + IV	FD	FD	FD + IV	FD + IV
<i>Change in total share of immigrants in farm employment between t and $t + 4$</i>	0.0132 (0.72)		−0.1901 (−0.51)		0.0365 (1.53)		−0.0053 (−0.02)	
<i>Change in share of immigrants between t and $t + 4$ from:</i>								
<i>Western Europe, USA, etc. (group 1)</i>		−0.0288 (−0.72)		0.2159 (0.07)		0.0409 (0.83)		−0.4922 (−1.36)
<i>Eastern Europe (group 2)</i>		0.0026 (0.11)		1.6885 (0.26)		0.0463 (1.53)		0.8771 (0.76)
<i>Other (LDC) countries (group 3)</i>		0.0860 (2.47)**		0.2810 (0.20)		0.0039 (0.08)		0.8717 (1.43)
Region $\times$ sub-sector $\times$ year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Farm controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	23,995	23,995	22,452	22,452	26,394	26,394	24,771	24,771
Number of establishments	5,766	5,766	5,239	5,239	6,643	6,643	6,060	6,060
R-squared	0.0814	0.0816	0.0737	−0.2269	0.1043	0.1043	0.1070	0.0414

Note: Estimations are based on a panel of individual native workers from 1980 to 2008, conditional on a worker being employed at the same farm in t and $t + 4$. Results above are for $t = 1984, 1988, 1992, 1996, 2000$, and 2004. Region $\times$ sub-sector $\times$ year dummies are constructed from three regions and four subsectors. The farm controls used are: Age of establishment at time t (25 categories), number of employees at time t (11 categories), average tenure at time t , and average experience at time t . IV estimations are implemented as 2SLS using the lagged share(s) of immigrants from $t - 2$ as instrument(s). 1st stage F-statistics are as follows: 5.8 (column 3), 7.4, 1.8, and 5.9 (column 4), 17.9 (column 7) and 15.7, 2.3, and 3.3 (column 8). t -values in parentheses. ***, ** and * indicate significance at the 1, 5, and 10% levels, respectively. Standard errors are clustered at the establishment level.

13% in 2008—an increase that is largely due to an influx of Eastern European immigrants. At the same time, the agricultural sectors of developed countries have continued their structural transformation, resulting in fewer but larger and presumably more efficient farms.

In this article, we have exploited very detailed matched employer-employee data for almost 30 years to analyze the farm-level relationship between immigrant labor and performance. We found that farms that employ immigrant workers are larger in terms of employees, revenue, and value added—and at least as productive as farms that do not use immigrants.

Our empirical analysis revealed that farms that increase their share of immigrants—especially immigrants from Eastern Europe and less developed countries—also tend to become larger in terms of employment and revenue. Furthermore, our instrumental-variable regressions indicated that this was not just due to a selection effect in the sense that growing farms find it easier to employ

immigrants, but that immigrants also enable farms to grow.

Our theoretical model indicated two potential explanations for this. Immigrants could constitute a cheaper/more flexible source of labor than native workers and hence reduce the marginal cost of labor for farms with access to immigrant labor, or they could bring in complementary skills that would boost productivity. The fact that immigrants from LDCs and Eastern Europe are the lowest-paid immigrants and the finding that the employment growth took place largely (or at least partly) at the expense of native labor without any effects on labor productivity both support the former explanation. This is also consistent with the observation that effects are short-lived and disappear when the immigrants leave. Finally, we found that immigrants tend mainly to substitute for unskilled natives, while the effects on skilled natives are much less negative (or even absent).

The findings of the present article indicate that immigrants may play an important role in the ongoing transformation of the agricultural

sector. Large and growing farms find it easier to use immigrants, and immigrants in turn reduce the farms' costs.

An issue that we have only briefly touched upon in the present article is the role of immigrant employers in the agricultural sector. These have also become more prevalent in recent years, and may find it even easier to make efficient use of immigrants in their production. However, that topic is left to future research.

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References

- Angrist, J.D., and A.D. Kugler. 2003. Protective or Counter Protective? Labour Market Institutions and the Effects of Immigration on EU Natives. *Economic Journal* 113 (488): F302–F331.
- Angrist, J.D., and J.-S. Pischke. 2009. *Mostly Harmless Econometrics*. Princeton, NJ: Princeton University Press.
- Aydemir, A., and G. Borjas. 2007. Cross-Country Variation in the Impact of International Migration: Canada, Mexico, and the United States. *Journal of the European Economic Association* 5 (4): 663–708.
- Borjas, G.J. 2003. The Labor Demand Curve is Downward Sloping: Re-examining the Impact of Immigration on the Labor Market. *Quarterly Journal of Economics* 118 (4): 1335–1374.
- Borjas, G.J. 2006. Native Internal Migration and the Labor Market Impact of Immigration. *Journal of Human Resources* 41 (2): 221–258.
- Borjas, G.J., R.B. Freeman, and L.F. Katz. 1997. How Much Do Immigration and Trade Affect Labor Market Outcomes? *Brookings Papers on Economic Activity* 1: 1–90.
- Card, D. 1990. The Impact of the Mariel Boatlift on the Miami Labour Market. *Industrial and Labour Relations Review* 43 (2): 245–257.
- Card, D. 2001. Immigrant Inflows, Native Outflows, and the Local Labor Market Impacts of Higher Immigration. *Journal of Labor Economics* 19 (1) 22–64.
- Card, D. 2005. Is the New Immigration Really So Bad? *Economic Journal* 115 (507): F300–F323.
- Cortes, P. 2008. The Effect of Low-Skilled Immigration on U.S. Prices: Evidence from CPI Data. *Journal of Political Economy* 116 (3): 381–422.
- Davis, S.J., and J.C. Haltiwanger. 1992. Gross Job Creation, Gross Job Destruction and Employment Reallocation. *Quarterly Journal of Economics* 107 (3): 819–863.
- Davis, S.J., J.C. Haltiwanger, and S. Schuh. 1996. *Job Creation and Destruction*. Cambridge, MA: MIT Press.
- Devadoss, S., and J. Luckstead. 2008. Contributions of Immigrant Farmworkers to California Vegetable Production. *Journal of Agricultural and Applied Economics* 40 (3): 879–894.
- Dustmann, C., F. Fabbri, and I. Preston. 2005. The Impact of Immigration on the British Labour Market. *Economic Journal* 115 (507): F324–F341.
- Huffman, W.E. 2005. Trends, Adjustments, and Demographics, and Income of Agricultural Workers. *Review of Agricultural Economics* 27 (3): 351–360.
- Kasimis, C. 2005. Migrants in the Rural Economies of Greece and Southern Europe. *Migration Information Source* October 2005.
- Longhi, S., P. Nijkamp, and J. Poot. 2005. A Meta-analytic Assessment of the Effects of Immigration on Wages. *Journal of Economic Surveys* 19 (3): 451–477.
- Longhi, S., P. Nijkamp, and J. Poot. 2006. The Impact of Immigration on the Employment of Natives in Regional Labour Markets: A Meta-analysis, IZA Discussion Paper No. 2044, Institute for the Study of Labor (IZA).
- Malchow-Møller, N., J.R. Munch, and J.R. Skaksen. 2012. Do Immigrants Affect Firm-Specific Wages? *Scandinavian Journal of Economics* 114 (4): 1267–1295.
- Martin, P.L., and J.E. Taylor. 1998. Poverty Amid Prosperity: Farm Employment, Immigration, and Poverty in California. *American Journal of Agricultural Economics* 80 (5): 1008–1014.
- Ottaviano, G.I.P., and Peri, G. 2012. Rethinking the Effects of Immigration on Wages. *Journal of the European Economic Association* 10 (1): 152–197.
- Ottaviano, G.I.P., G. Peri, and G.C. Wright. 2012. Immigration, Offshoring and

- American Jobs. *American Economic Review*, forthcoming.
- Partridge, M.D., D.S. Rickman, and K. Ali. 2008. Recent Immigration and Economic Outcomes in Rural America. *American Journal of Agricultural Economics* 90 (5): 1326–1333.
- Pischke, J.-S., and J. Velling. 1997. Employment Effects of Immigration to Germany: An Analysis Based on Local Labor Markets. *Review of Economics and Statistics* 79 (4): 594–604.
- Staiger, D., and J.H. Stock. 1997. Instrumental Variables Regression with Weak Instruments. *Econometrica* 65 (3): 557–586.
- Statistics Denmark. 2001. *50-års oversigten*. Copenhagen, Denmark: Danmarks Statistik.
- Stock, J.H., and M. Yogo. 2002. Testing for Weak Instruments in Linear IV Regression. National Bureau of Economic Research (NBER) Technical Working Paper 284.
- Taylor, J.E., and P.L. Martin. 1997. The Immigrant Subsidy in US Agriculture: Farm Employment, Poverty and Welfare. *Population and Development Review* 23 (4): 855–874.
- Taylor, J.E., and P.L. Martin. 2001. Human Capital: Migration and Rural Population Change. In *Handbook of agricultural economics*, vol. 1, part 1, ed. B.L. Gardner and G.C. Rausser, 457–511. New York: Elsevier Science.
- Taylor, J.E., and P.L. Martin. 2003. Farm Employment, Immigration, and Poverty: A Structural Analysis. *Journal of Agricultural and Resource Economics* 28 (2): 349–363.
- Venturini, A. 1999. Do Immigrants Working Illegally Reduce the Natives' Legal Employment? Evidence from Italy. *Journal of Population Economics* 12 (1): 135–154.
- Wooldridge, J.M. 2002. *Econometric analysis of cross-section and panel data*. Cambridge, MA: MIT Press.